

AERO 540: Advanced Aerospace Propulsion and Rocketry

Term: Fall 2026 | Instructor: Prof. Robert Goddard (rgoddard@aero.univ.edu)

Course Description & Overview:

An advanced engineering study of liquid and solid chemical rocket propulsion systems, supersonic combustion ramjets, and electric ion propulsion thrusters.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
Rocket Combustion Chamber Design	25%	Module 04	CAD & Thermal Engineering Report
Supersonic Nozzle CFD Simulation	25%	Module 08	OpenFOAM Mesh & Whitepaper
Propulsion System Flight Test Capstone	35%	Module 11	Static Fire Test Video & Dossier
Engineering Seminar Technical Participation	15%	Continuous	Bi-Weekly Lab Critique

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: Thermodynamics of Rocket Propulsion	Sutton (Ch. 1–3); NASA SP-8087 Specs
Week 02	Module 02: Chemical Propellant Combustion Dynamics	Sutton (Ch. 4 & 5); Liquid Oxygen Engine Manual
Week 03	Module 03: De Laval Nozzle Flow & Shock Waves	Anderson (Ch. 3–5)
Week 04	Module 04: Turbopump Cycles & Injector Design	Huzel & Huang (Ch. 6); Turbopump Specs
Week 05	Module 05: Cryogenic Propellant Management	NASA Cryogenic Fluid Guidelines
Week 06	Module 06: Combustion Instability & Acoustic Waves	Harrje & Reardon (1972)
Week 07	Module 07: Solid Rocket Motor Grain Geometries	Sutton (Ch. 11–13)
Week 08	Module 08: Electric & Ion Propulsion Systems	Jahn & Choueiri (2002); NEXT Thruster Specs
Week 09	Module 09: Hypersonic Scramjet Propulsion	Heiser & Pratt (Ch. 4); AIAA Whitepaper
Week 10	Module 10: Reusable Booster Thermal Protection	FAA Launch Safety Standards
Week 11	Module 11: Flight Test Showcase & Demos	Industry Propulsion Architecture Whitepapers